BCS-404	SOFTWARE ENGINEERING	L T P 3 1 0	4 credits

Pre-requisites: Basic computer knowledge

Course Objectives:

CO1	to apply engineering and computer science concepts in the development and maintenance of reliable, usable, and dependable software
CO2	To understand the nature of software development and software life cycle process models, agile software development.
CO3	To learn concepts and principles in parallel with the software development life cycle.
CO4	To know basics of testing and understanding concept of software quality assurance and software configuration management process.
CO5	To understand project scheduling concept and risk management associated to various type of projects.

Detailed Syllabus

MODULE-I

Introduction: Introduction to Software Engineering, Software Components, Software Characteristics, Software Crisis, Software Engineering Processes, and Software Quality Attributes. Software Development Life Cycle (SDLC) Models: Water Fall Model, Prototype Model, Spiral Model, Evolutionary Development Models, Iterative Enhancement Model. Software Requirement Specifications (SRS) Requirement Engineering Process: Elicitation, Analysis, Documentation and Review, Feasibility Study, Information Modeling, Data Flow Diagrams, Entity Relationship Diagrams, Decision Tables, SRS Document, IEEE Standards for SRS. Software Quality Assurance (SQA): Verification and Validation, SQA Plans, Software Quality Frameworks, ISO 9000 Models, SEI-CMM Model.

MODULE-II

Software Design: Basic Concept of Software Design, Architectural Design, Low Level Design: Modularization, Design Structure Charts, Pseudo Codes, Flow Charts, Coupling and Cohesion Measures, Design Strategies: Function Oriented Design, Object Oriented Design, Top-Down and Bottom-Up Design. Software Measurement and Metrics: Various Size Oriented Measures: Halestead's Software Science, Function Point (FP) Based Measures, cyclomatic Complexity Measures: Control Flow Graphs. Software Testing: Testing Objectives, Unit Testing, Integration Testing, Acceptance Testing, Regression Testing, Testing for Functionality and Testing for Performance, Top-Down and BottomUp Testing Strategies: Test Drivers and Test Stubs, White Box Testing, Black Box Testing, Alpha and Beta Testing of Products. Static Testing Strategies: Formal Technical Reviews (Peer Reviews), Walk Through, Code Inspection, Compliance with Design and Coding Standards.

MODULE-III

Software Maintenance and Software Project Management: Software as an Evolutionary Entity, Need for Maintenance, Categories of Maintenance: Preventive, Corrective and Perfective Maintenance, Cost of Maintenance, Software Re- Engineering, Reverse Engineering. Software Configuration Management Activities, Change Control Process, Software Version Control, An Overview of CASE Tools. Estimation of Various Parameters such as Cost, Efforts, Schedule/Duration, Constructive Cost Models (COCOMO), Resource Allocation Models, Software Risk Analysis and Management.

Text Books:-

1. R. S. Pressman, Software Engineering: A Practitioners Approach, McGraw Hill.
2. Rajib Mall, Fundamentals of Software Engineering, PHI Publication.

Reference Books:-

1. K. K. Aggarwal and Yogesh Singh, Software Engineering, New Age International Publishers.
2. PankajJalote, Software Engineering, Wiley
3. Carlo Ghezzi, M. Jarayeri, D. Manodrioli, Fundamentals of Software Engineering, PHI Publication.

Course Outcomes: After the completion of the course the student will be able to:

CO1	To understand basic concept of software engineering, different phases to make a software & study them in detail, project management concepts & their metrics, design models & its principles
CO2	Discuss requirement engineering and its models (Information, functional, behavioural), different testing techniques for different projects
CO3	Implement Software life cycle models
CO4	Compare different types of models
CO5	Calculation of staffing for a particular project, its cost & schedule